

RDM-7 Visual Designer — User Guide

1. Getting Started

What is the RDM-7 Visual Designer?

The RDM-7 Visual Designer is a desktop application for designing custom automotive dashboard and gauge cluster layouts for RDM-7 display hardware. Build dashboards visually, map live CAN bus signals, and push layouts directly to your device.

Supported Displays

Display	Resolution	Shape
RDM-7 inch	800 x 480	Rectangle
4 inch Round	720 x 720	Circle
3 inch Round	800 x 800	Circle
2.8 inch Round	480 x 480	Circle

First Launch

When you open the app for the first time, a **Getting Started Tour** will walk you through the main areas of the interface. You can replay it anytime from the menu.

The app works in **offline mode** by default — you can design layouts without a device connected. Connect later to push your work to the RDM-7.

2. The Editor

Layout

The editor is divided into three main areas:

- **Toolbar** (top) — Display size selector, screen mode (Dashboard / Splash), layout selector, connection controls, and Save button. — **Canvas** (centre) — The live preview of your dashboard. This is where you drag, position, and resize widgets. — **Right Sidebar** — Three tabs: — **Widgets** — Palette of available widget types to drag onto the canvas. — **Properties** — Inspector for the selected widget's configuration. — **Layers** — Z-order list of all widgets on the layout.

Canvas Controls

Action	How
Move widget	Click and drag
Resize widget	Select, then drag a corner or edge handle

Action	How
Multi-select	Ctrl+Click widgets, or drag a selection box on empty canvas
Pan canvas	Hold Space + drag
Zoom	Ctrl + scroll wheel
Zoom to fit	Click the reset button in the toolbar
Clone widget	Alt + drag
Nudge 1px	Arrow keys
Nudge 10px	Shift + Arrow keys

Alignment Tools

The alignment bar appears above the canvas:

- **Align Left / Centre / Right** — Snap selected widgets horizontally. — **Align Top / Middle / Bottom** — Snap selected widgets vertically. — **Distribute X / Y** — Evenly space 3 or more selected widgets.

Smart Snapping

When dragging widgets, guide lines appear automatically when edges or centres align with other widgets. The snap threshold is 10px.

Toolbar Toggles

- **Bezel** — Show/hide the device frame around the canvas. — **Widgets** — Show/hide widget selection borders (toggle off for a clean preview). — **Crosshair** — Show/hide alignment crosshair guides. — **Sim** — Enable signal simulation for testing without a device.

3. Building a Dashboard

Adding Widgets

1. Open the **Widgets** tab in the right sidebar. 2. Drag a widget type onto the canvas. 3. Release to place it. The widget appears at the drop position.

Configuring a Widget

1. Click a widget to select it. 2. Switch to the **Properties** tab. 3. Adjust fields like label, font, colours, decimals, min/max values, and more. 4. Changes appear in the live canvas preview immediately.

Widget Types

Display Widgets

Widget	Purpose
Panel	Numeric value readout with label and unit. Supports alert colour thresholds.

Widget	Purpose
Bar Graph	Horizontal fill bar with min/max scaling and colour alerts.
RPM Bar	Full-width RPM bar with redline marker and limiter flash effects (7 modes).
Meter	Circular analogue gauge with needle, tick marks, and scale labels.
Arc	Partial arc gauge with fill animation. Supports image overlays for custom designs.
Shift Light	LED strip that fills with RPM. Configurable LED count, colours, and flash point.
Text / Value	Static text label or live signal value. Custom font, colour, and rotation.
Gear Display	Shows the current gear selection.

Visual Widgets

Widget	Purpose
Image	Display a PNG/JPG from your image library. Scale 10-200%, tint overlay.
Shape Panel	Decorative rectangle with background, border, corner radius, and shadow.
Indicator	Turn signal indicator (left/right) with animation.
Warning Light	Circular alert icon. Activates when a signal crosses a threshold.

Interactive Widgets (CAN Transmit)

Widget	Purpose
Toggle Switch	Tap to send a CAN value. Supports latching and custom graphics.
Button	Momentary or latching CAN-transmit button. Configurable TX rate and colours.

Conditional Rules

Most widgets support **rules** — automatic appearance changes when a signal meets a condition.

Example: Turn a panel's background red when coolant temperature exceeds 100 degrees.

To add a rule: 1. Select the widget. 2. Open Properties, scroll to **Rules**. 3. Click **Add Rule**. 4. Set the condition (signal, operator, threshold). 5. Set the override (which property changes, and to what value).

Rules are evaluated in real-time on the device.

Layer Order

Widgets are drawn in layer order — widgets higher in the Layers tab appear on top. To change order: — **Right-click** a widget and choose **Bring Forward** or **Send Back**. — Or drag widgets in the **Layers** tab to reorder.

4. Signals & Data

What Are Signals?

Signals map CAN bus messages to values your widgets display. Each signal defines how to extract a number from a raw CAN frame:

Field	Description
Name	Human-readable identifier (e.g. "RPM", "COOLANT_TEMP")
CAN ID	Message identifier (hex, e.g. 0x520)
Bit Start	Position of the first bit in the message (0-63)
Bit Length	Number of bits to extract (1-32)
Scale	Multiply raw value by this (e.g. 0.1 means raw 1000 = display 100.0)
Offset	Add this after scaling
Endian	Byte order — Little Endian (Intel) or Big Endian (Motorola)
Signed	Whether the value can be negative

ECU Presets

The fastest way to get started is with an ECU preset:

1. Select your ECU from the dropdown in the toolbar (e.g. **MaxxECU**). 2. All standard signals are loaded automatically. 3. When configuring a widget, pick a signal from the dropdown — they're pre-configured with the correct CAN IDs, bit positions, and scaling.

DBC File Import

If your ECU provides a .dbc file (CAN database):

1. Open the Signal Manager (Ctrl+Shift+S). 2. Click **Import DBC**. 3. Select your .dbc file. 4. Signals are extracted and added to your layout.

Custom Signals

For signals not covered by presets:

1. Open the Signal Manager (Ctrl+Shift+S). 2. Click **Add Signal**. 3. Fill in the CAN ID, bit position, length, scale, and offset. 4. The signal is now available in widget property dropdowns.

Assigning Signals to Widgets

1. Select a widget on the canvas. 2. Open the **Properties** tab. 3. Find the **Signal** field. 4. Pick from the dropdown or type to search.

5. Advanced Features

Splash Screens

The splash screen is displayed briefly when the RDM-7 boots up.

1. Switch to **Splash Screen** mode using the dropdown in the toolbar. 2. A default splash with the RDM logo is provided. 3. Add images, text, and shapes — the same widget tools apply. 4. Save to push the splash to the device. 5. Enable **Fade** for a smooth transition from splash to dashboard.

Signal Simulation

Test your layout without a connected device:

1. Click **Sim** in the toolbar to enable simulation. 2. Click the gear icon next to it for simulation settings. 3. Simulated values animate across your widgets so you can verify thresholds, colours, and behaviour.

Images & Fonts

Upload custom images and fonts via the **Storage Manager** (accessible from the menu):

- **Images**: PNG or JPG files. Used in Image widgets, gauge backgrounds, needle overlays. — **Fonts**: TTF files. Used in Text, Panel, and other widgets with text.

Data Logger

Record live CAN signal data:

1. Open **Data Logger** from the menu. 2. Click **Start** to begin recording. 3. Click **Stop** when done. 4. Download the CSV file for analysis in a spreadsheet.

Fuel Calibration

Calibrate your fuel tank level sensor:

1. Open **Fuel Calibration** from the menu. 2. With the tank empty, click **Set Empty**. 3. Fill the tank, then click **Set Full**. 4. The RDM-7 interpolates between these reference points.

6. Device Management

Connecting to Your Device

Mode	Setup	Best For
USB Serial	Plug in USB cable. Port auto-detected.	File transfers, firmware updates
WiFi	Enter device IP from the connection dropdown.	Live preview, data logging
ESP32 Hotspot	Connect PC to the RDM-7's WiFi network.	Track-side use, no router needed
Offline	No connection needed.	Designing without hardware

Saving to Device

When connected, **Save** (Ctrl+S) pushes your layout directly to the device's internal storage. The display updates in real time.

Storage Manager

Open from the menu to manage files on the device:

- **Internal Storage** — Layouts, images, and fonts stored on the RDM-7. — **SD Card** — Backup and transfer. Copy files between internal and SD card. — View storage usage and free space.

Firmware Updates (OTA)

1. Open the menu and select **Update Firmware (OTA)**.
2. The app checks for the latest firmware version.
3. Follow the prompts to download and flash the update over USB or WiFi.

Import / Export Layouts

- **Export Layout** — Saves a .rdm file to your computer for sharing or backup. — **Import Layout** — Load a .rdm file into the editor (Ctrl+O).

7. Quick Reference

Keyboard Shortcuts

Shortcut	Action
Ctrl+S	Save layout
Ctrl+O	Import RDM file
Ctrl+Z / Ctrl+Y	Undo / Redo
Ctrl+C / Ctrl+V	Copy / Paste widget
Ctrl+D	Duplicate widget
Ctrl+Shift+C / V	Copy / Paste style
Delete	Delete widget
Arrow keys	Nudge 1px
Shift+Arrows	Nudge 10px
Ctrl+Scroll	Zoom in/out
Space+Drag	Pan canvas
Alt+Drag	Clone widget
Ctrl+Click	Multi-select
Right-Click	Context menu
Ctrl+Shift+S	Signal Manager

Display Presets

Preset	Resolution	Shape
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Troubleshooting

Problem	Solution
Can't connect via USB	Check the COM port is correct. Try unplugging and replugging the cable.
Can't connect via WiFi	Verify the IP address. Ensure your PC and the RDM-7 are on the same network.
Widgets not updating	Check that signals are assigned and the device is connected.
Layout looks different on device	Ensure the screen preset matches your physical display.
SD card not detected	Check the card is inserted and formatted as FAT32.